PHASE 1 · DATASET VALIDATION · CFM

State Continued Claims

The dataset, its clocks, and every validation check

Weekly state-level continued & initial claims from FRED/ALFRED · source notebook: claims_validation_new.ipynb

51	**2,113**	**84,248**	**41,718**	**25**
states	weeks of history	archived first prints	point-in-time rows	validation checks

Registered results: **12 PASS** **6 WARN** **0 FAIL**

July 2026

One report, two series: how state claims reach the world

- **What it is** — `{ABBR}CCLAIMS` / `{ABBR}ICLAIMS`: state-level **continued** and **initial** unemployment-insurance claims. Weekly, **not seasonally adjusted**, 50 states + DC, served by **FRED** with archival vintages in **ALFRED**.
- **H1 · Publication schedule** — released every **Thursday 8:30 AM ET** in the DOL's *Unemployment Insurance Weekly Claims* report; moved to **Wednesday** when Thursday is a federal holiday. Verified in V3.2 (96.7% Thu + 3.3% Wed) and V3.3.
- **H2 · Reference period** — the reference week ends on the **Saturday two weeks before** the release Thursday, so a continued-claims number first appears at **obs + 12 days**. Continued claims trail initial claims by one week (you certify for the *prior* week). Verified in V2.1 and V3.3.
- **Print vocabulary** — the **advance print** is the first, preliminary figure for a week (Thursday right after the certification week); the **revised print** replaces it one report later. Which one FRED stores changed over time — the single most important quirk in this dataset (V3.7).

Note — **DOL = U.S. Department of Labor.** Its Employment & Training Administration (ETA) runs the UI system, which is why the report is often called the “DOL ETA” or “ETA 539” report. States transmit their counts to DOL; FRED/ALFRED then archives what DOL publishes — two hand-offs, each with its own timing fingerprint (V3.4, V3.5).

Three tables, three clocks — same numbers, different questions

table	file	rows	key (grain)	span	question it answers
vintage (cc_v)	claims_vintage_ first_release.csv	84,248 = 42,124 CC + 42,124 IC	(state, claim_type, obs_date)	obs 2010-08-28 → 2026-06-27 released 2010-09-16 → 2026-07-07	“What was **first published** for week t, and **when**?”
current (cc_c)	claims_current_ history.csv	214,909 = 107,452 CC + 107,457 IC	(state, claim_type, obs_date)	obs 1986-01-04 → 2026-06-27	“What does FRED say **today** about week t?” (fully revised)
pit	claims_pit_ features.csv	41,718 = 51 states × 818 Fridays	(as_of_date, state)	as-of 2010-11-05 → 2026-07-03	“What did we actually **know on Friday d**?” (backtest feed)

Note — The same physical number lives in all three tables — what differs is **when you are allowed to see it**. \`cc_v\` runs on the **information clock** (what the world learned, and when), \`cc_c\` on the **observation clock** (the cleanest history, for targets and seasonality), and \`pit\` replays both clocks so that **a backtest can only touch what was known at the time**. Mixing them up is how look-ahead bias happens.

Column dictionary — vintage & current tables

column	table	meaning
state / abbr / fips	both	full state name, USPS code, FIPS code (51 incl. DC)
claim_type	both	'cclaims' (continued) or 'iclaims' (initial) — the two series stacked in one long table
obs_date	both	reference **week-ending Saturday** — the observation clock (V2.1: 100% Saturdays)
value_current	current	today's fully-revised value for that week
first_release_date	vintage	day the value **first entered the ALFRED archive** — the information clock
value_first_release	vintage	the first print — what a real-time observer saw
is_current	vintage	True if the first print still equals today's value (CC: **96.7%** of rows — V4.1)
pub_lag_days	vintage	first_release_date – obs_date. CC: **13–26d**, median 19d — two regimes, see V3.7

Note — \`is_current\` and \`pub_lag_days\` are **derived** convenience columns — everything reduces to the two date stamps plus the value. If you remember one thing: **obs_date says what the number measures; first_release_date says when anyone could know it.**

Column dictionary — the point-in-time panel (30 columns)

column(s)	meaning
as_of_date	the **vantage point**: a Friday; every cell in the row reflects only data released **on or before** this date (818 consecutive Fridays)
state	one row per state per Friday — always balanced 51/51 (V1.1)
cclaims_t0 … t7 / iclaims_t0 … t7	the **8 most recent weekly readings available** as of that Friday; t0 = newest. Staggered capture never creates holes — it shows up as an older t0
cclaims_latest_obs / iclaims_latest_obs	which observation week t0 refers to
cclaims_lag_days / iclaims_lag_days	as_of_date – latest_obs: how **stale** t0 is. IC runs ~7 days fresher than CC (one-week reporting offset, H2)
*_4wma, *_4wma_prev, *_wow_pct, *_4wma_pct	4-week moving average, its previous value, week-over-week % and 4wma momentum — computed **only from released readings**

Discussion — **"Vantage point" in plain English:** as_of_date is the day you pretend it is *now*. The builder replays history — it stands on each Friday, looks backward through the release calendar, and writes down only what had been published by then. That is why the panel has **zero missing cells and zero look-ahead** (V3.9): a late-arriving state just carries a bigger lag_days, never a hole.

Why the three tables start on different dates

layer	starts	why it can't start earlier
current history	**1986-01-04**	FRED serves today's vintage across the whole history — no point-in-time obligation, so it reaches back as far as the series exists
vintage (first prints)	**2010-08-28** (first release 2010-09-16)	ALFRED only began **archiving snapshots** of state claims in Sep-2010. A first print that was never snapshotted cannot be recovered — the boundary is an archive fact, not a data defect (V3.1)
pit panel	**2010-11-05**	a row needs a full stack of 8 weekly readings (t0…t7) before it exists → first valid Friday ≈ 7 weeks after the vintage boundary

- The ordering **current** ≥ **vintage** ≥ **pit** is structural — it held identically for state U3 and NFP (1976/1990 → 2005 → 2006).
- Consequence: **1986 → 2010-09 is NOT point-in-time.** Usable for seasonality and climatology; never for backtesting (V3.1).

Note — Reading `describe()` output: the **50% row is the midpoint of the sample, not a boundary**. The pit panel's median as-of lands mid-2018 simply because the panel spans 2010→2026 — the *start* is the min row. (The same confusion came up for U3, where the median 2017 was mistaken for the start.)

25 checks in 4 modules — how to read the rest of this deck

module	theme	checks	registered results
V1	Structural integrity — states, keys, nulls, values, zeros, truncation	V1.0 – V1.5 (+ missing-value heatmaps)	print-style flags (OK / WARN)
V2	Calendar & frequency — the **observation clock** (the week being measured)	V2.1, V2.2a, V2.2b, V2.3	3 PASS · 1 WARN
V3	Point-in-time & release calendar — the **information clock** (when it became known)	V3.1 – V3.9 (incl. V3.5b)	7 PASS · 3 WARN
V4	Revision behavior — first release and after	V4.1, V4.1b, V4.2, V4.3	2 PASS · 2 WARN

- **PASS** = criterion met · **WARN** = usable with a documented caveat · **FAIL** = blocker. This run: **12 PASS, 6 WARN, 0 FAIL.**
- Every check states its **question and criterion up front**; each slide that follows shows the check's evidence, verdict, and — where it matters — a short discussion of what it means for modeling.
- Scope: this notebook is the per-question restructure of modules **V1–V4**; deeper modules (outliers, coherence, seasonality, lead-lag, backtest-readiness, governance: V5–V10) live in the full suite `claims_validation.ipynb`.

MODULE

V1

Is the panel physically sound — right states, unique keys, complete fields, credible values?

Structural integrity

check	question it answers
V1.0	What do the three tables actually look like? (eyeball before testing)
V1.1	Do we have all 51 states in every table, every period?
V1.2	Is (state, claim_type, week) a unique key — no double-loaded rows?
V1.3	Are the key fields complete (no nulls)?
V1.4	Is every value parseable, finite and positive — across FRED's int→float format drift?
V1.4b	Are the zeros benign ramp-in artifacts or mid-sample reporting dropouts? Plus: what do the missing-value heatmaps really show?
V1.5	Did any state's fetch silently truncate?

V1.0 — The three tables at a glance

INFO

Criterion: look before testing — a validation suite that never prints its data validates blind.

- **vintage** (cclaims slice): 42,124 rows · first prints **132 → 4,808,361** — the max is California's May-2020 COVID peak, a useful sanity anchor · pub_lag 13–26d, median 19d.
- **current** (cclaims slice): 107,452 rows · obs **1986-01-04 → 2026-06-20** · 15 zero values, all ≤ 1986-07-19 (V1.4's subject).
- **pit**: 41,718 rows · **51 states × 818 Fridays** · 30 columns of look-ahead-free features.

	rows	value range	dates
vintage	42,124	132 .. 4,808,361	2010-08-28 .. 2026-06-20
current	107,452	0 .. 4,808,361	1986-01-04 .. 2026-06-20
pit	41,718	(t0..t7 features)	2010-11-05 .. 2026-07-03

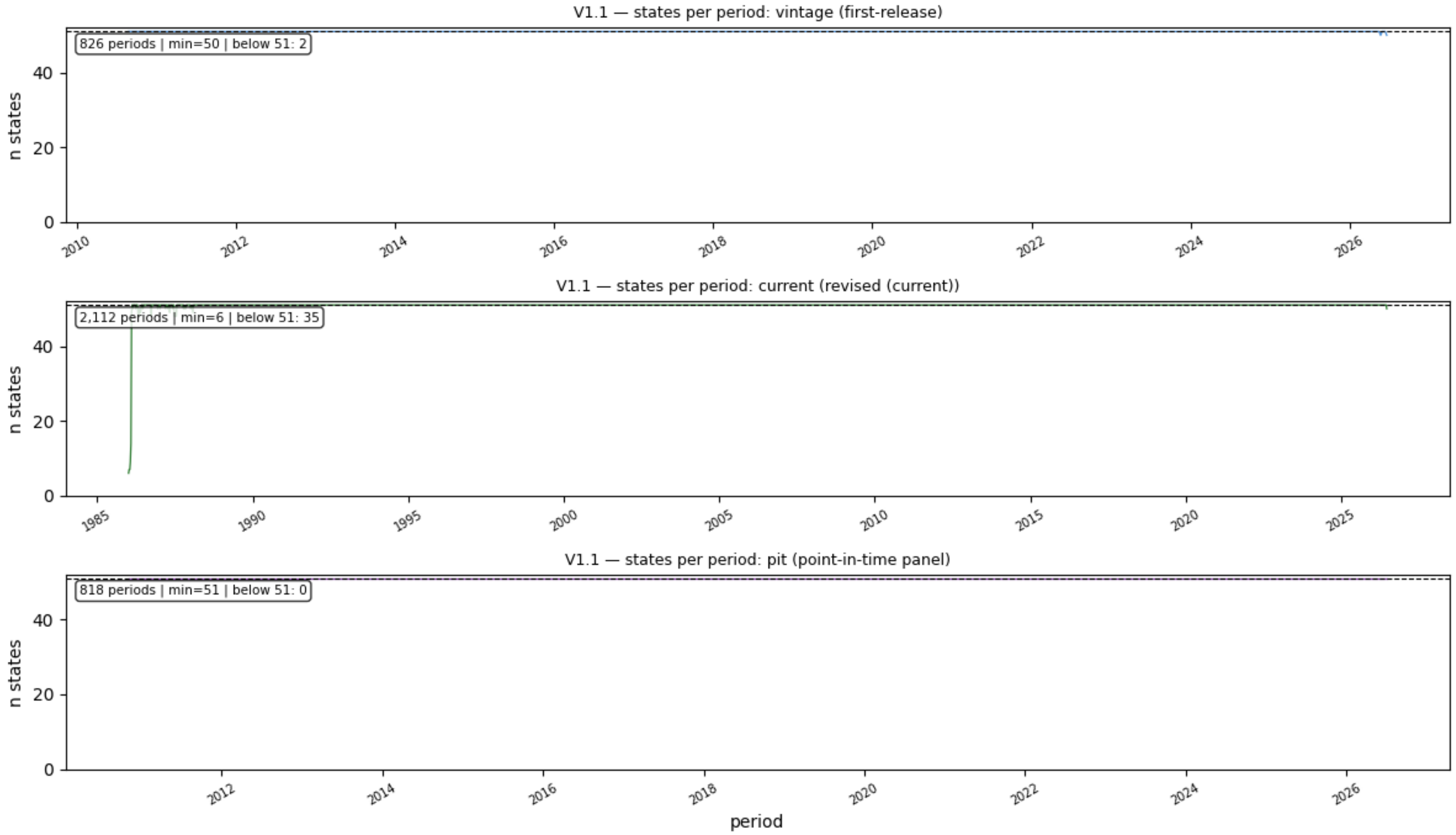
Note — Values span **4 orders of magnitude** across states and 36,000× within the sample — every downstream transform should be in logs or %, and cross-state comparisons need normalization (labor-force share).

V1.1 — Do we have all 51 states in every table, every period?

OK

Criterion: 51 unique states per table; 51 per period outside documented edges.

- Unique states: **vintage 51 · current 51 · pit 51.** Per period: vintage min 50 (**2 trailing weeks**), current min 6 (**1986 ramp-in**, 35 weeks below 51), pit **51 always**.



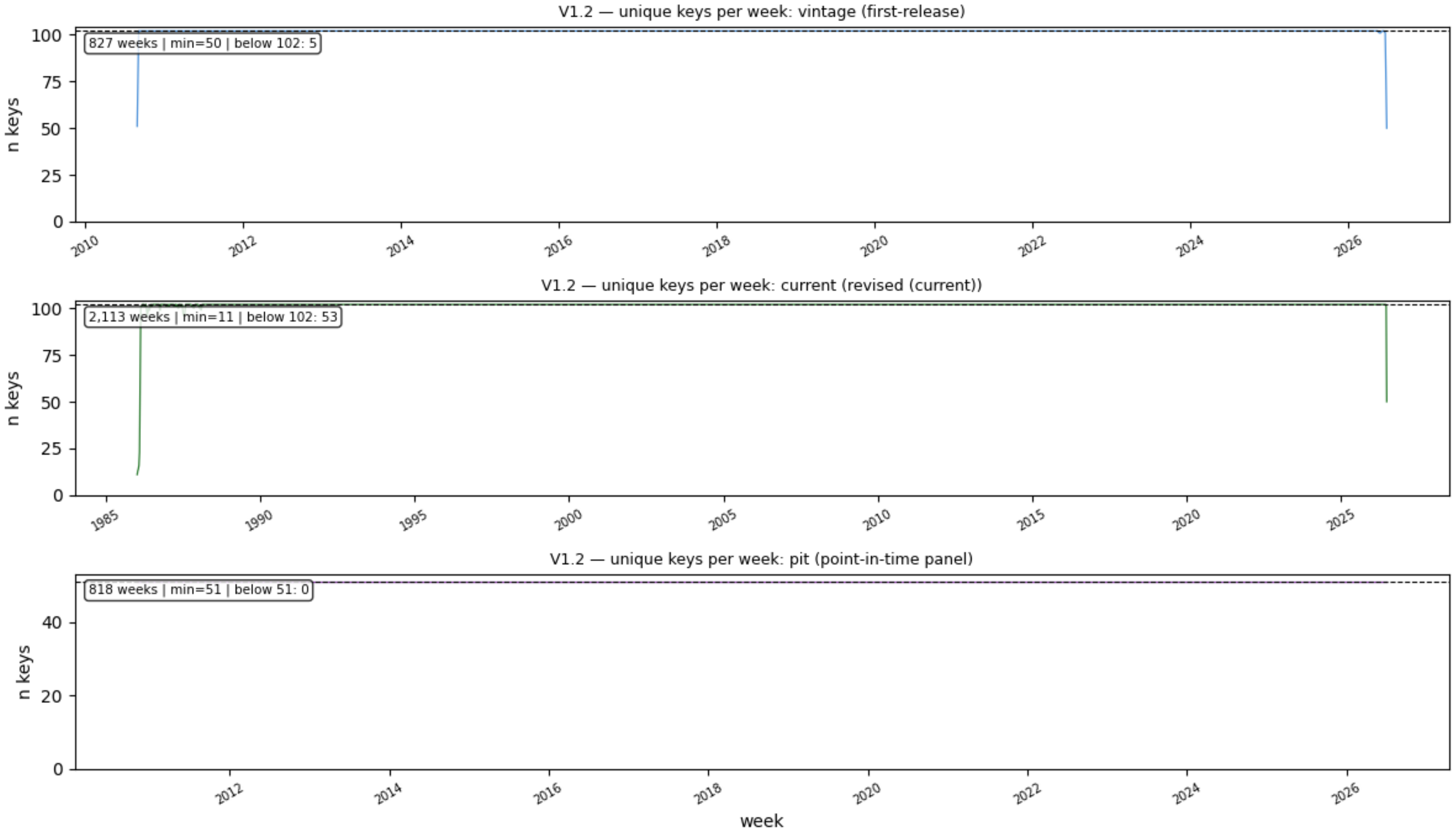
Note — The two vintage weeks at 50 are the **live capture edge** — the newest week where one state's print hadn't been archived yet at fetch time; they fill in at the next report. The 1986 dip is the startup ramp (V2.2a). The pit panel is balanced **by construction** — staggering becomes lag, not absence.

V1.2 — Is (state, claim_type, week) a unique key — no double-loaded rows?

OK

Criterion: 0 duplicate rows on the declared key of each table.

- **vintage 84,248 rows = 84,248 unique keys · current 214,909 = 214,909 · pit 41,718 = 41,718** (key: as_of_date × state) — zero duplicates anywhere.
- Keys per week: expected 102 (51 states × 2 claim types); vintage avg **101.9**, current avg **101.7**, pit exactly **51** — the below-expected weeks are the same edges as V1.1.



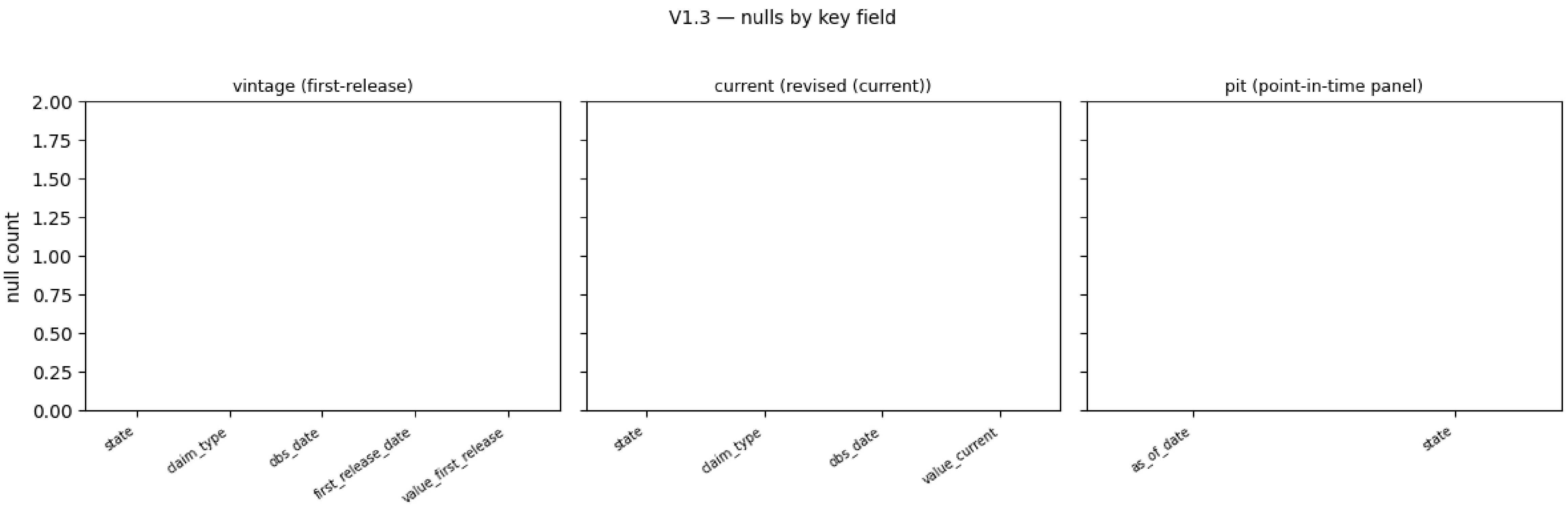
Note — Duplicate keys are the classic symptom of a fetch retried and appended twice. Zero duplicates + V1.5's row-count check together rule out both double-loading and silent truncation.

V1.3 — Are the key fields complete (no nulls)?

OK

Criterion: 0 nulls in every check field of all three tables.

- Field-by-field: **0 nulls** in state, claim_type, obs_date, first_release_date, value_first_release (vintage); value_current (current); as_of_date, state (pit).



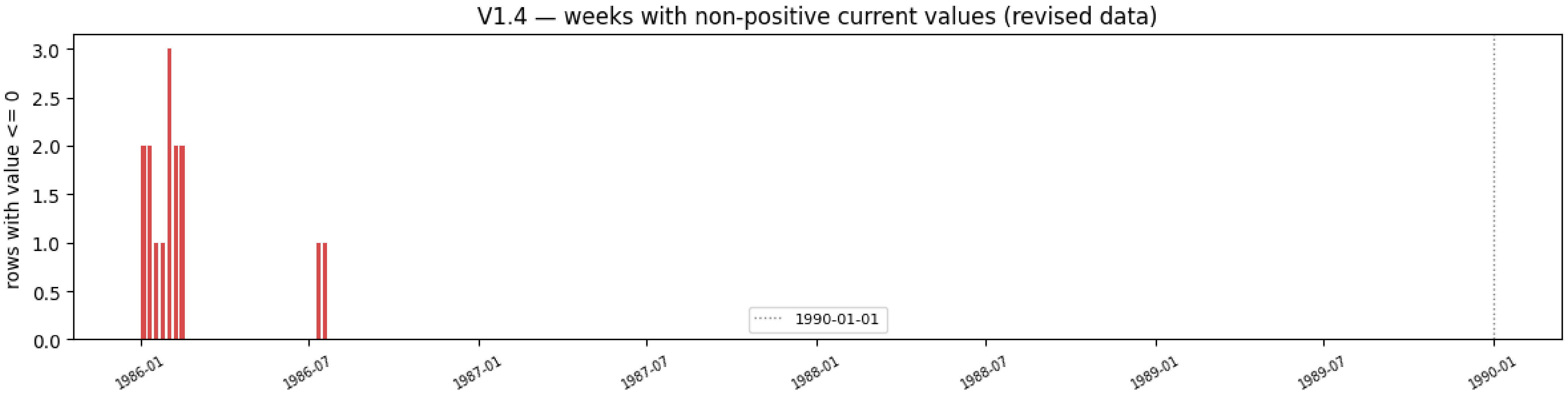
Note — FRED encodes a missing week as the string '.' — the fetch layer converts and drops those before assembly, so a null here would mean a parser regression, not bad source data. Genuinely unreported weeks appear as **absent rows**, which is V2.2's subject, not nulls.

V1.4 — Is every value parseable, finite and positive?

WARN

Criterion: 0 bad rows; startup-era zeros tolerated as WARN.

- vintage: **0** non-finite / non-positive rows. current: **15 zeros** — Colorado, Iowa, New Jersey, New Mexico, Virginia — the latest dated **1986-07-19**; **post-1990: 0**.
- FRED's mid-series **int→float format drift** (“345525” → “345525.0”) is absorbed by numeric coercion — a parse failure would surface here as non-finite.



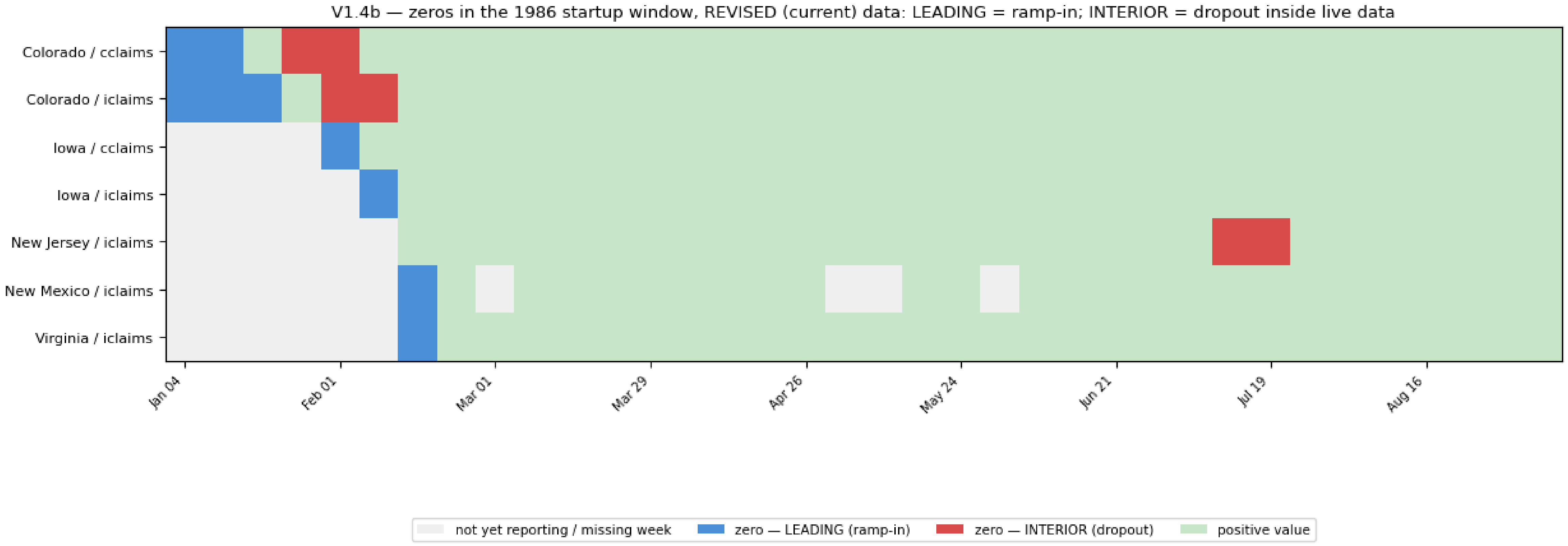
Note — A state with literally zero continued claims does not happen in a 2,100-week panel — so a zero is a **reporting artifact by definition**. All 15 sit in the 1986 startup window; V1.4b classifies them.

V1.4b — Are the zeros ramp-in artifacts (leading) or reporting dropouts (interior)?

WARN

Criterion: interior zeros == 0 ideally; pre-1990 occurrences tolerated as WARN.

- **9 leading** zeros (placeholder rows before a state starts reporting) vs **6 interior** (dropouts inside live data: CO ×4, NJ ×2) — **post-1990 interior: 0**.
- All 15 zeros live in the 1986 startup window (map below: blue = leading, red = interior, gray = not yet reporting).



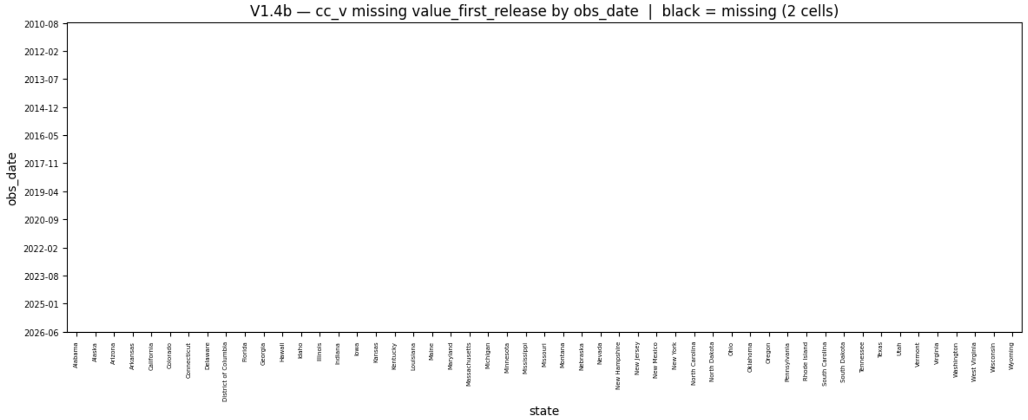
Note — Treat pre-1990 zeros as **missing, not as "zero claims"** — include them in the same NaN handling as the V2.2a gaps. In the modeling era (post-2010 vintages) there is not a single zero anywhere.

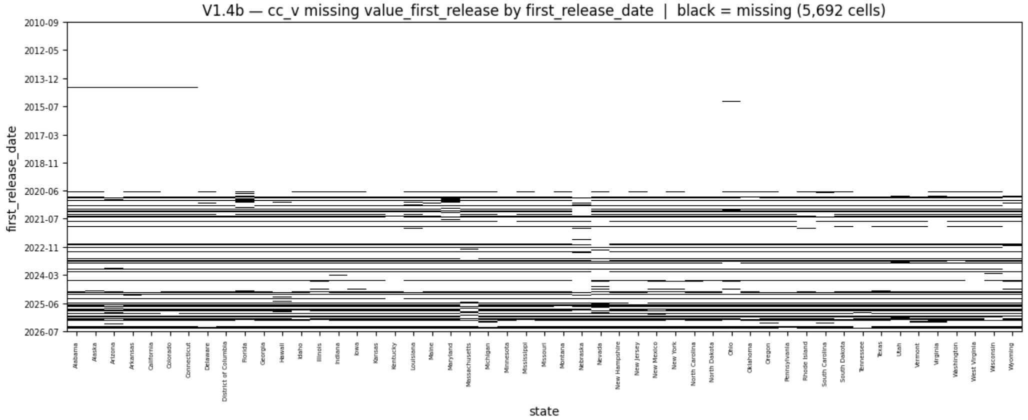
V1.4b (cont.) — Missing-value heatmaps — three pivots, three answers

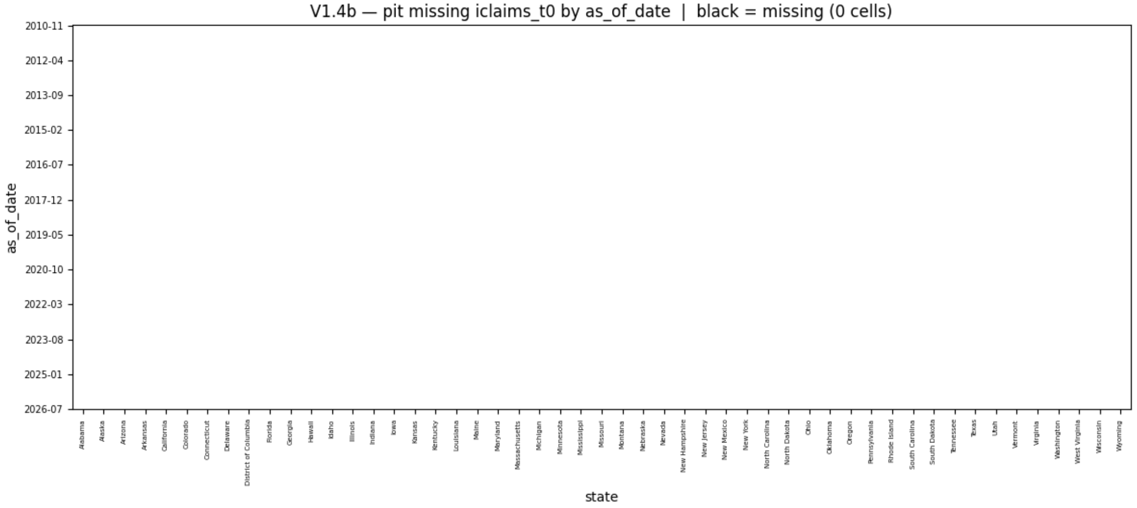
INFO

Criterion: the count of “missing” cells depends on the axis you pivot on; know which one you're reading.

- Same vintage table, three pivots: by **obs_date** → **2 missing** of 42,126 (trailing capture edge) · by **first_release_date** → **5,692 “missing”** of 47,481 (12%!) · pit by **as_of_date** → **0 missing** of 41,718.







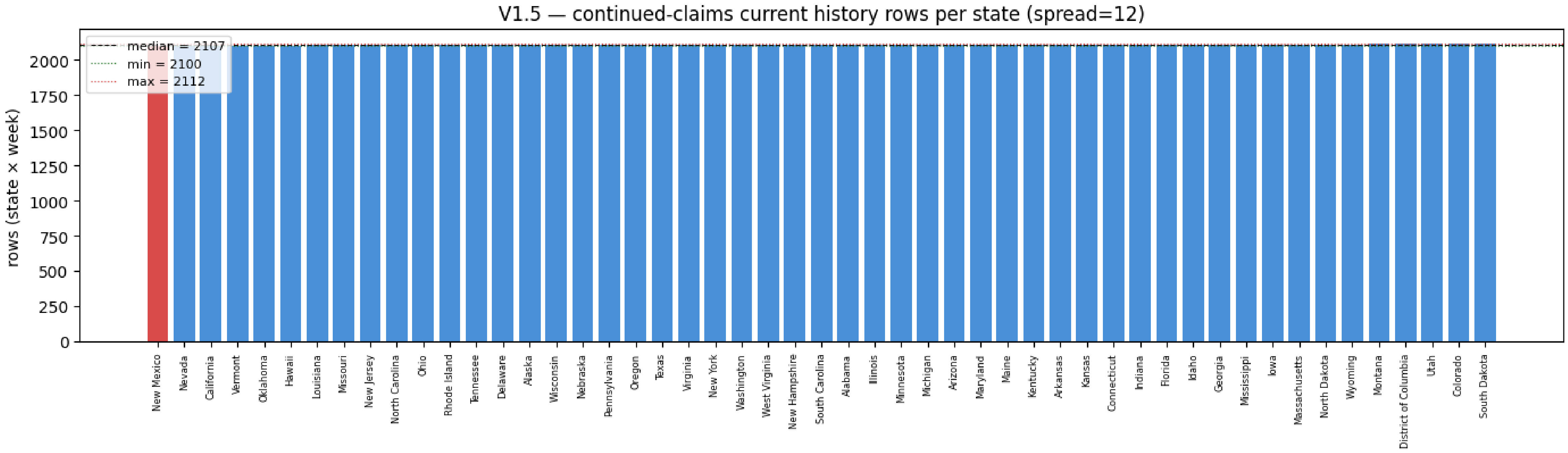
Note — first_release_date is **not a shared axis**: after Jul-2020 states are captured on different days, so each release date holds only the states captured that day — every other cell renders black. That is **staggering, not data loss**. The pit panel is indexed on a shared Friday grid and stores “latest available”, so staggering becomes **lag** (an older t0), never a hole.

V1.5 — Did any state's fetch silently truncate?

OK

Criterion: max – min rows per state ≤ 15 weeks.

- Rows per state (CC current): **min 2,100 · median 2,107 · max 2,112** — spread **12 ≤ 15**. Thinnest: New Mexico (–7 vs median); no state is hundreds of rows short.



Note — The residual spread traces to the ragged **1986 starts** (V2.3) plus each state's pre-1990 gaps — not to the API. A truncated fetch (FRED paginates at 100k observations) would show a state conspicuously short; none is.

MODULE

V2

The observation clock: obs_date stamps the week being measured — always a DOL Saturday.

Calendar & frequency

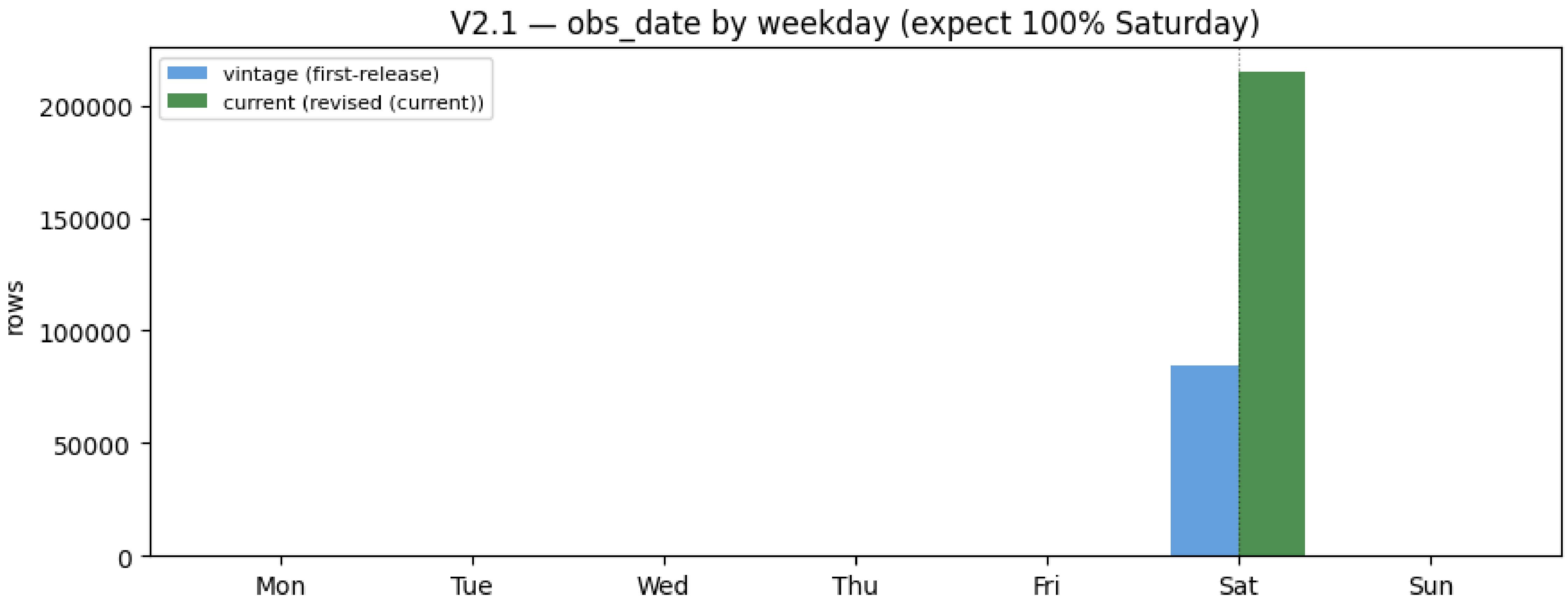
check	question it answers
V2.1	Do all observation dates follow the DOL Saturday week-ending convention?
V2.2a	Are there missing weeks in the old (pre-1990) history?
V2.2b	Are there missing weeks in modern data — live issues to monitor?
V2.3	Do all states cover the same window (balanced panel)?

V2.1 — Do all observation dates follow the DOL Saturday week-ending convention?

PASS

Criterion: 0 non-Saturday obs_date rows.

- **vintage: 827 unique weeks, all Saturday. current: 2,113 unique weeks, all Saturday.** Non-Saturday rows: **0** in both.



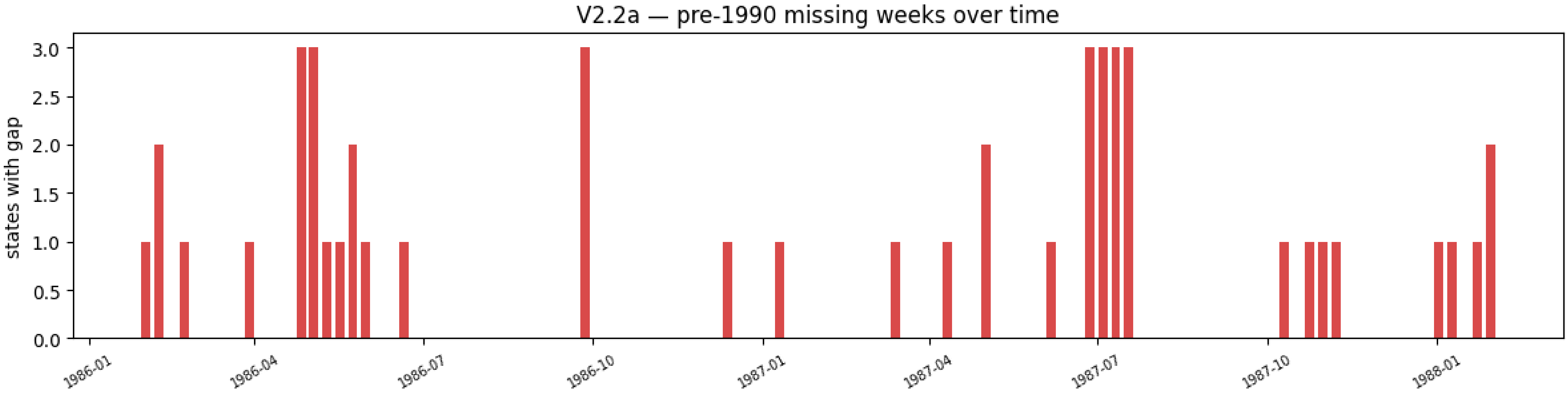
Note — H2 anchors here: the reference week ends the Saturday **two weeks before** the release Thursday. Saturday + 12 days = Thursday — which is exactly why the national first print lands at obs+12d (V3.3). Any monthly aggregation must respect these Saturday boundaries.

V2.2a — Are there missing weeks in the old (pre-1990) history?

WARN

Criterion: 0 ideally; startup-era holes tolerated as WARN.

- **48 missing interior weeks across 22 states, all 1986–88** — worst: New Mexico 7, Nevada 5, California 5. Early DOL collection holes from the program's electronic-reporting ramp-up.



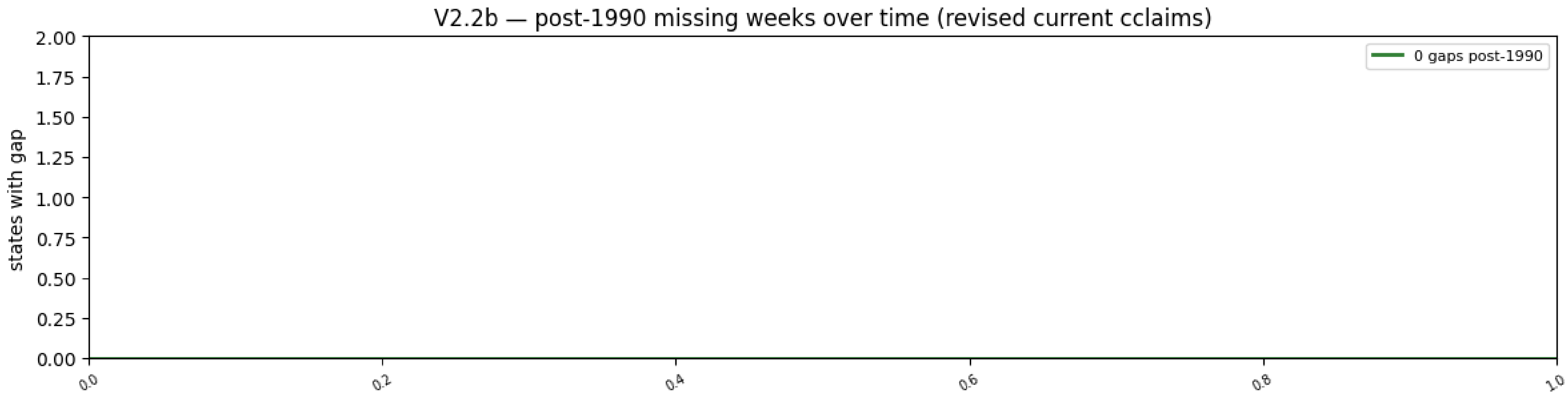
Note — RULE for pipelines: **do not forward-fill through a gap** when computing week-over-week changes — a WoW across a hole is a two-week change in disguise. Pre-1990 history is for seasonality estimation only, and estimators must tolerate NaN weeks.

V2.2b — Are there missing weeks in modern data — live issues to monitor?

PASS

Criterion: == 0; any occurrence must be investigated.

- Post-1990 interior gaps: **0 missing weeks across 0 states.** Thirty-six years of a 51-state weekly panel with no holes.



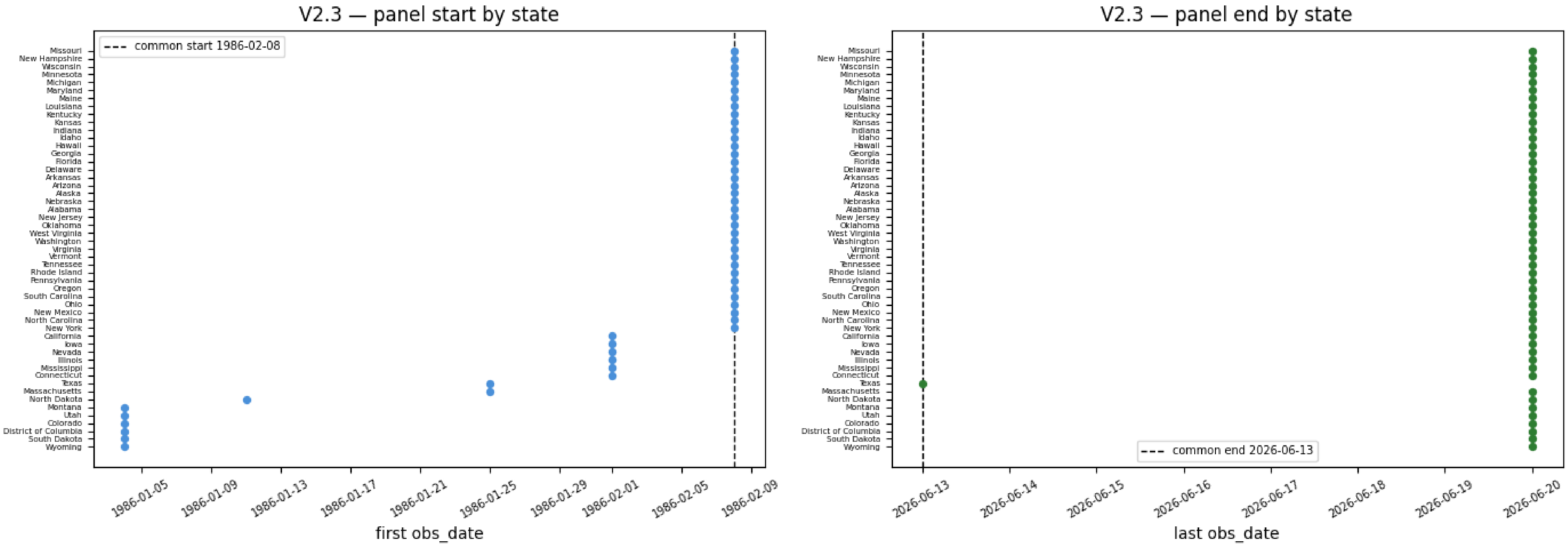
Note — If a gap ever appears here it means a state failed to report (FRED shows '.') — monitor for backfill. The pit builder needs no special case: the availability frontier simply skips the unreported week, and t0 goes stale by 7 extra days.

V2.3 — Do all states cover the same window (balanced panel)?

PASS

Criterion: start spread ≤ 10 weeks; end spread ≤ 1 week.

- Start spread **5 weeks** (1986-01-04 → 1986-02-08) · end spread **1 week** (2026-06-13 → 2026-06-20). Common balanced window: **1986-02-08 → 2026-06-13 = 2,106 weeks**.



Note — The 1-week end spread is the **live boundary**: one state's newest week simply hadn't been captured yet on fetch day — it closes at the next Thursday report. Anyone requiring a strictly balanced panel should clip to the common window.

MODULE

V3

The information clock: first_release_date stamps when the number became known. This module is what makes backtesting legal.

Point-in-time & release calendar

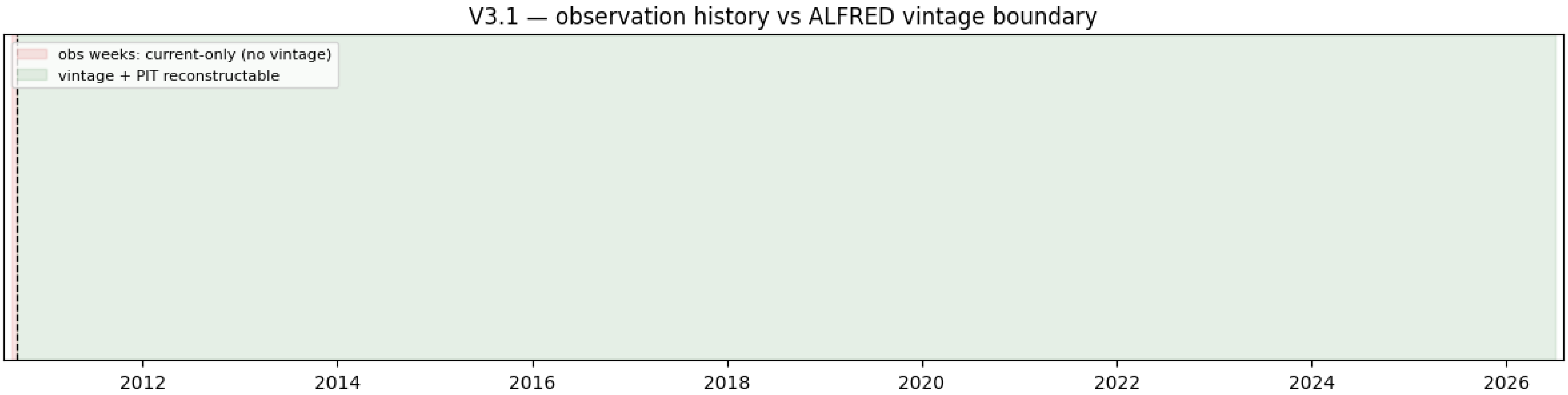
check	question it answers
V3.1	How far back can point-in-time be reconstructed?
V3.2	Does the vintage calendar match the DOL mid-week release schedule?
V3.3	Is the national first release the advance print at obs+12d?
V3.4	On which weekday does FRED capture state data?
V3.5	How long after the public release do state values appear?
V3.5b	Is the state-vs-national gap pattern stable within each era?
V3.6	Could a backtest have seen state data before it was public?
V3.7	Has FRED's capture timing / stored print type been stable?
V3.8	Do weeks become available in order?
V3.9	Does the PIT panel contain any look-ahead?

V3.1 — How far back can point-in-time be reconstructed?

WARN

Criterion: boundary documented (WARN = limitation, not defect).

- First ALFRED vintage: **2010-09-16 — all 51 states archived from day one.** PIT panel reconstructable from **2010-11-05**; last vintage 2026-07-07.
- **153 vintage rows** carry obs weeks from before the boundary (history visible in the first snapshot) — current-only, not point-in-time.



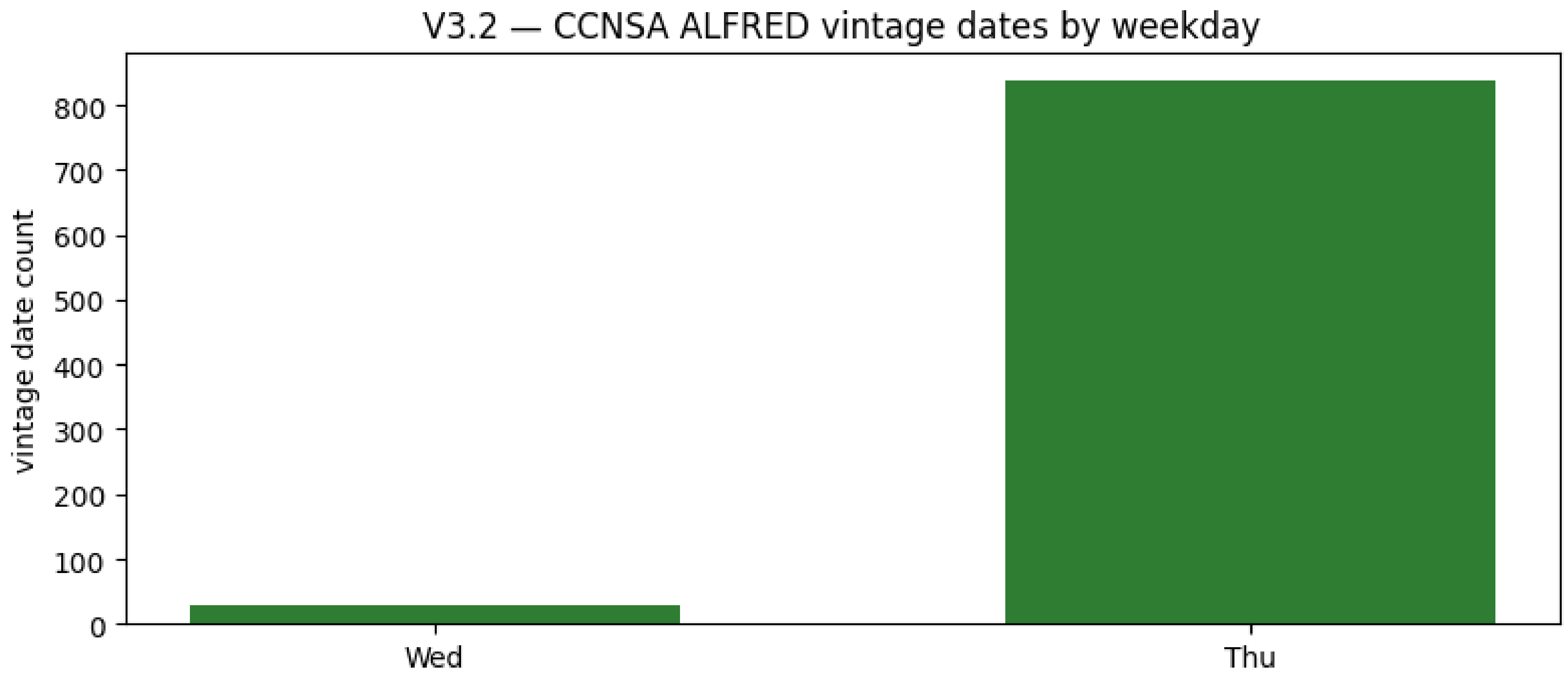
Note — Everything left of the red line exists only as **today's values**: 24 years (1986→2010) usable for seasonality and climatology, **never for backtesting**. This single date splits the dataset's two lives.

V3.2 — Does the vintage calendar match the DOL mid-week release schedule?

PASS

Criterion: Thu + Wed ≥ 99% of national vintage dates.

- 867 national CCNSA vintage dates: **Thursday 838 (96.7%) + Wednesday 29 (3.3%) = 100%.** Not one Monday, Tuesday, Friday or weekend vintage.



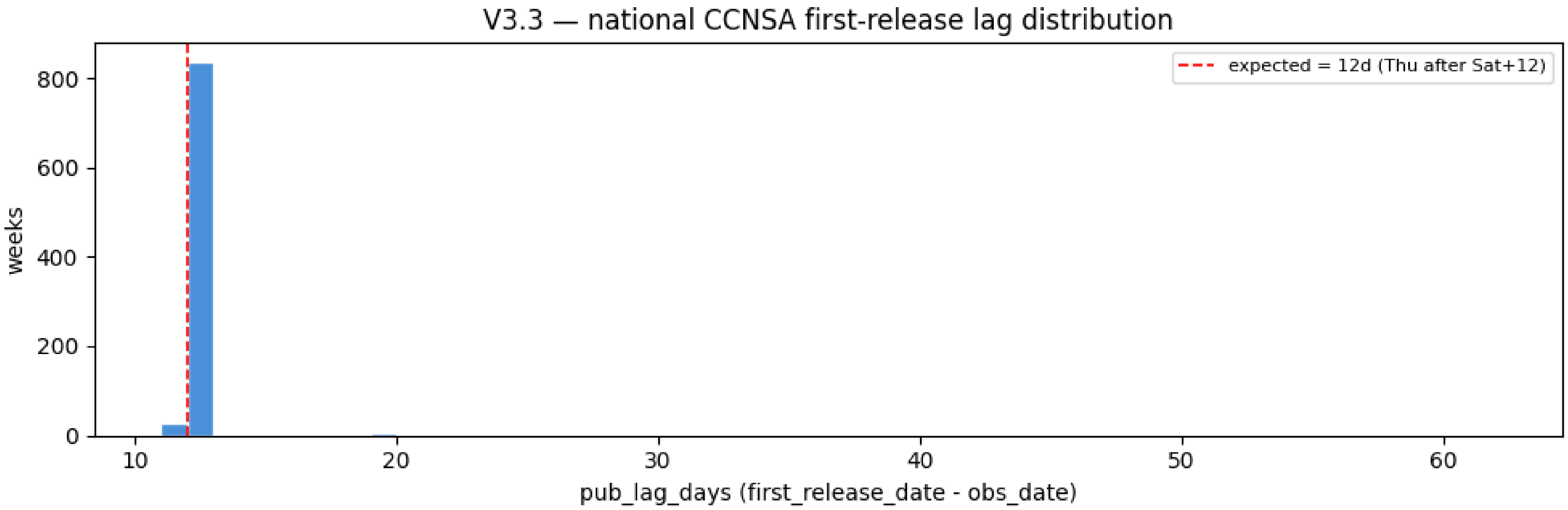
Note — The 29 Wednesdays are the **holiday-shifted reports** (Thanksgiving, July-4 weeks…) — DOL moves the release a day earlier, and ALFRED faithfully records it. The mid-week hypothesis H1 is confirmed against the archive, not against documentation.

V3.3 — Is the national first release the advance print at obs+12d?

PASS

Criterion: median publication lag == 12 days.

- 877 national first releases: **median = mode = 12 days** (Saturday week-end + 12d = report Thursday). Range 11–61d.
- The 40 exceptions decompose cleanly: **11d = holiday-Wednesday** reports; **19–61d = ALFRED archiving a handful of weeks late** (capture artifact, not DOL timing).



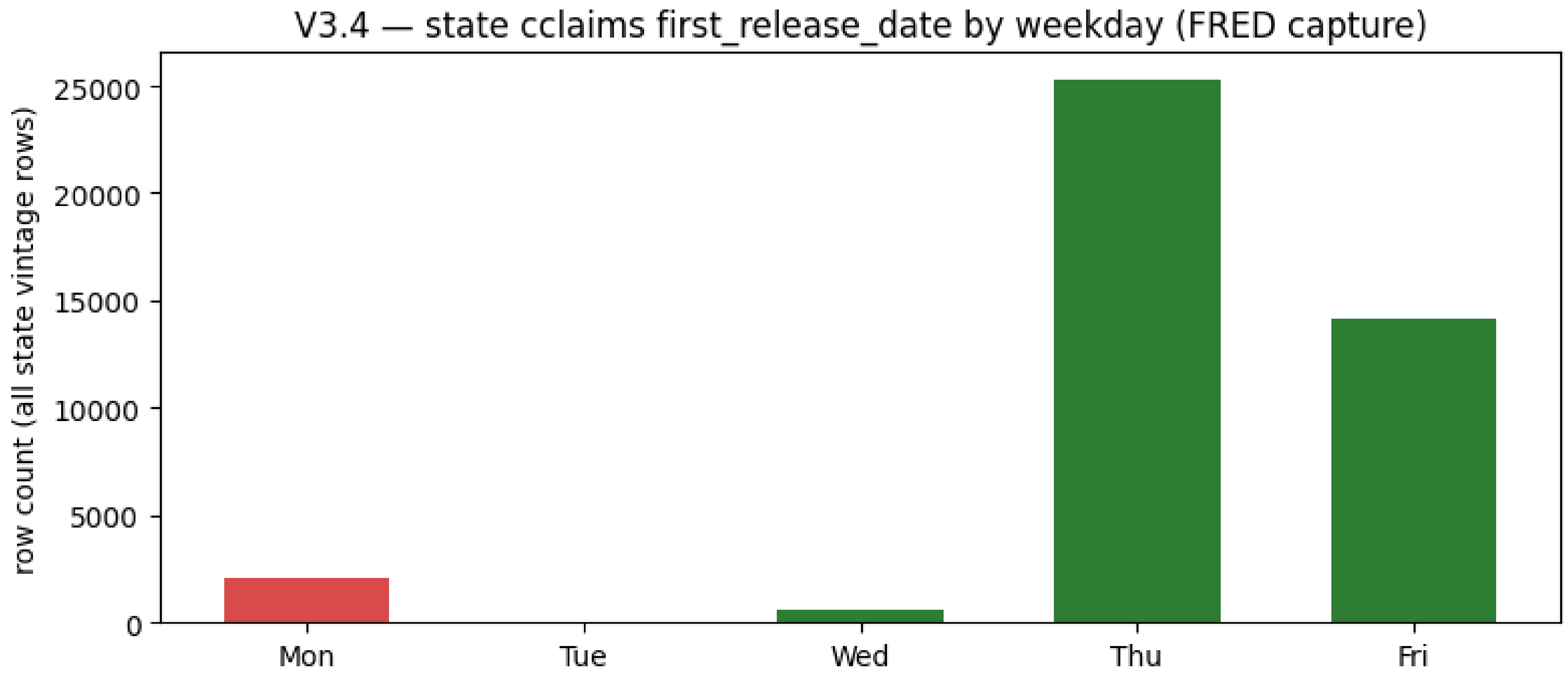
Note — obs+12d **is** the advance print. Continued claims sit one week behind initial claims in the same report (IC appears at obs+5d) — the certification lag of H2, and the reason `iclaims_t0` runs ~7 days fresher in the pit panel.

V3.4 — On which weekday does FRED capture state data?

PASS

Criterion: Thu + Fri + Wed ≥ 90% of state capture dates.

- 42,124 state first releases: **Thu 60.1% + Fri 33.5% + Wed 1.3% = 94.9%**; Monday 4.9%, Tuesday 0.1%.
- Thursday captures = the pre-2020 era archiving on the following **report day** (the revised print); Friday = the post-2020 era ingesting the **advance print next morning** (V3.7 explains the split).



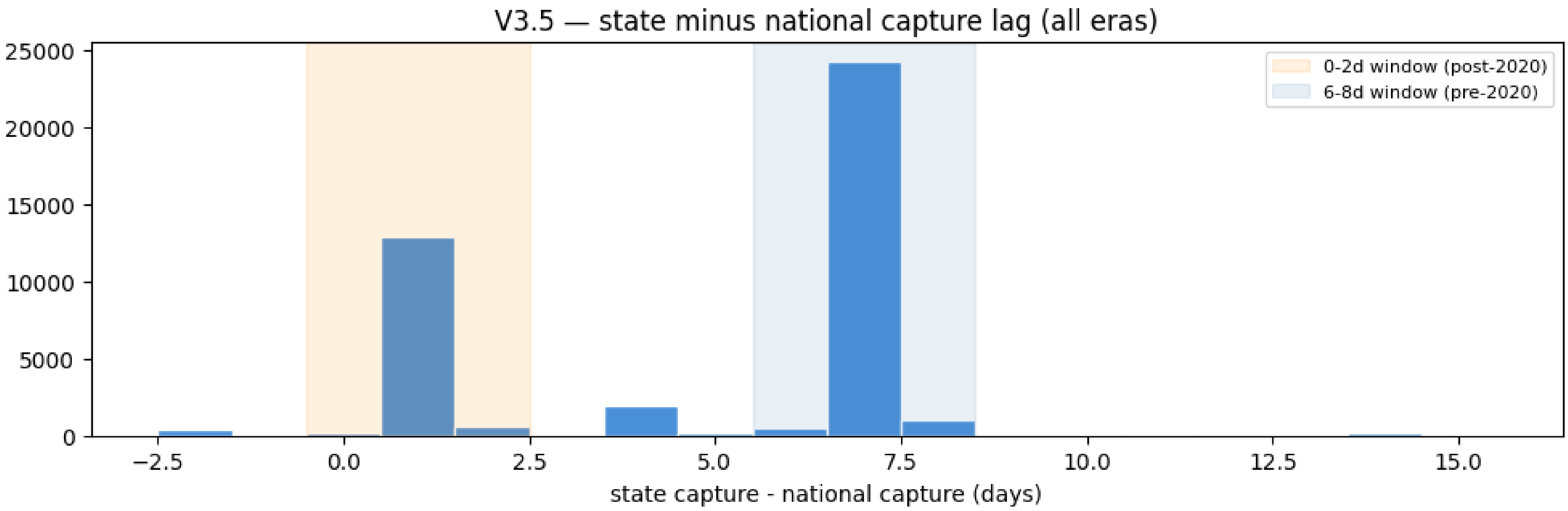
Note — The Monday tail is **FRED's weekend backlog**, not DOL timing — a release that reached the archive 3–4 days late. It is measurement noise on the capture side, quantified per era in V3.5b.

V3.5 — How long after the public release do state values appear?

PASS

Criterion: ≥ 90% inside one of the two report windows (0–2d or 6–8d).

- Gap between state capture and the national release of the same obs week: **61.1% at +6–8d** (pre-2020 pattern: the *next* report) · **32.5% at 0–2d** (post-2020: next day) · together 93.6%.
- Worked examples — obs 2015-03-07: national Thu 03-19, CA/TX/NY all **+7d**; obs 2023-03-04: national Thu 03-16, all three **+1d**.



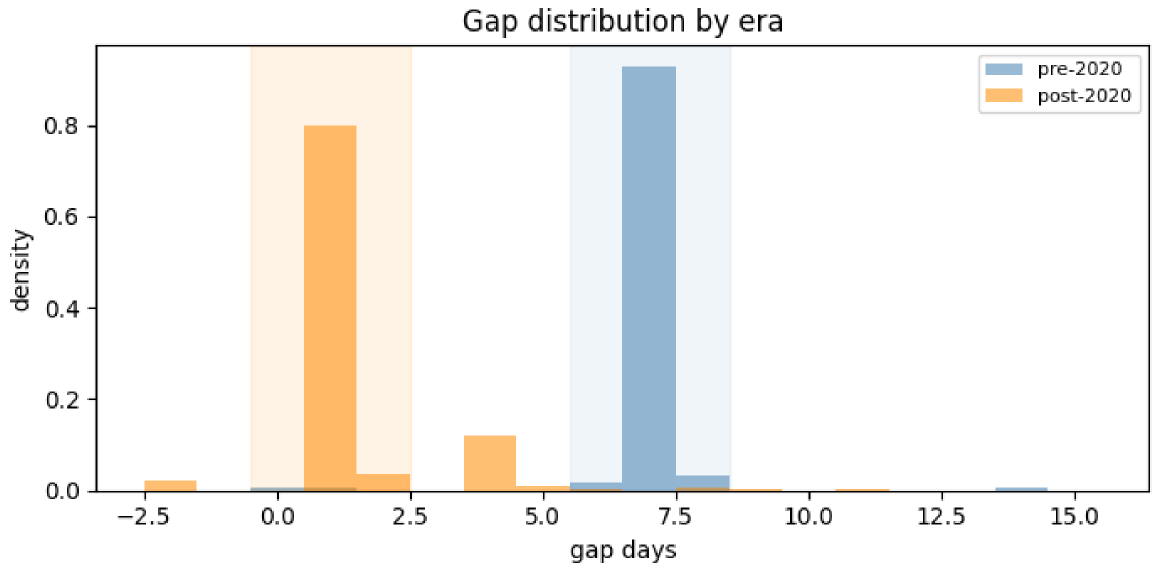
Note — This measures **ALFRED capture order, not DOL publication order** — the state numbers were public in the same Thursday report; FRED just archived them later. ALFRED-based PIT is therefore **conservative, never optimistic** — the safe direction for a backtest. For lead-lag work you may re-time availability to the DOL calendar.

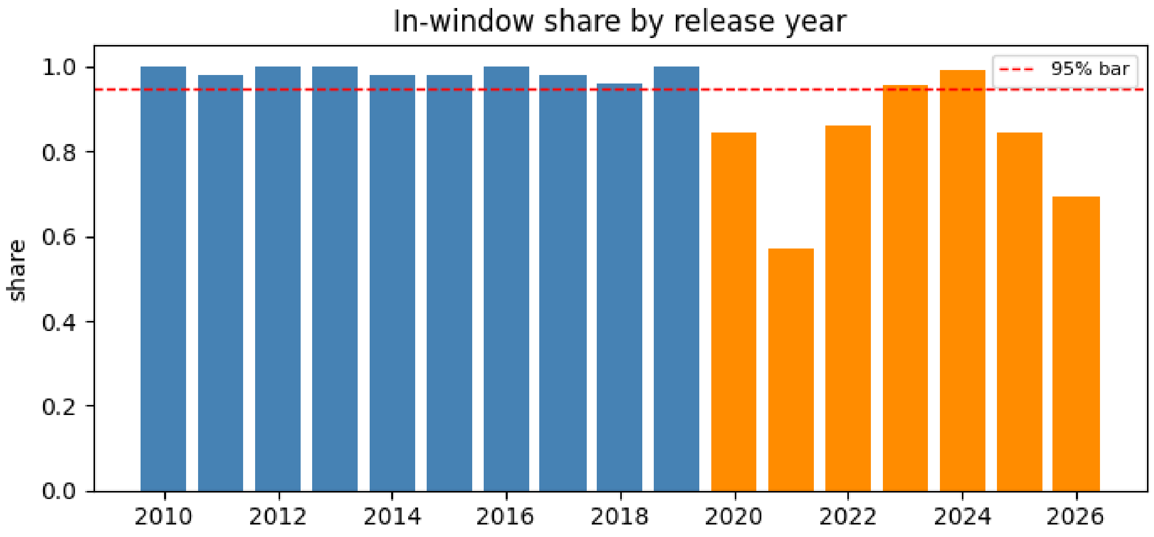
V3.5b — Is the state-vs-national gap pattern stable within each era?

WARN

Criterion: ≥ 95% in-window per era.

- Pre-2020: **97.8%** at +6–8d — near-perfect discipline. Post-2020: **83.3%** at +0–2d with **13.0% slipping to +3–5d** (the Monday ingestion tail); 2021 is the chaotic transition year (57%).





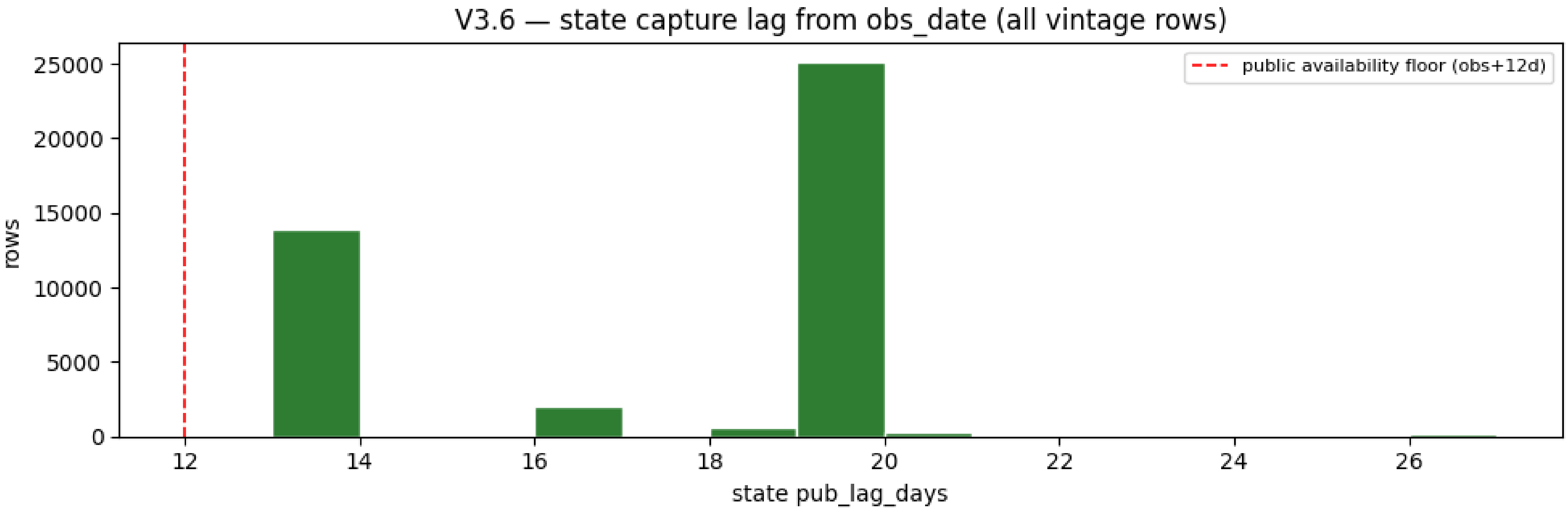
Note — The post-2020 imperfection is FRED ops, bounded at about **+4 days**. Practical rule: treat post-2020 availability as "release Thursday + up to 4 days" in the worst case — or re-time state features to the DOL calendar if the 1–4-day jitter matters for your horizon.

V3.6 — Could a backtest have seen state data before it was public?

PASS

Criterion: minimum state publication lag ≥ 12 days (the DOL floor).

- Minimum state pub_lag across all 42,124 vintage rows: **13 days** — one day *above* the 12-day floor; rows below 12d: **0**.
- A naive “state archived before national” comparison fires **357 false alarms on 7 weeks** — every one is a late *national* archive (e.g. obs 2025-09-20: national at obs+61d, states at obs+13d).



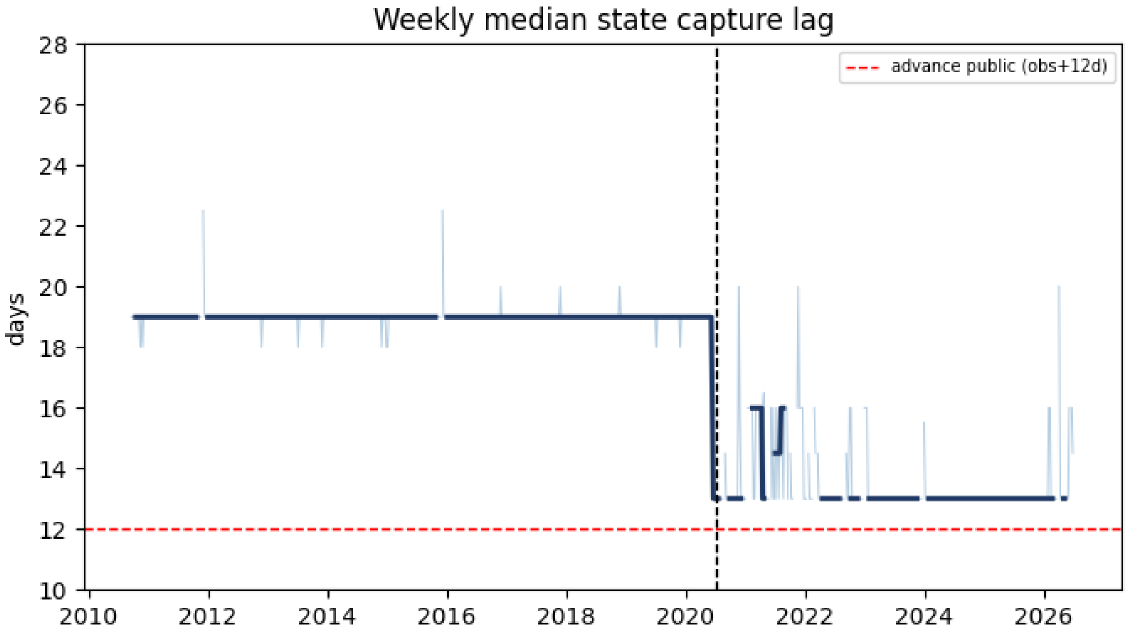
Note — Leak tests must compare against the **DOL calendar**, not against another ALFRED capture date — otherwise the reference itself can be late and the test cries wolf. Against the true floor, there is no leak anywhere in 16 years.

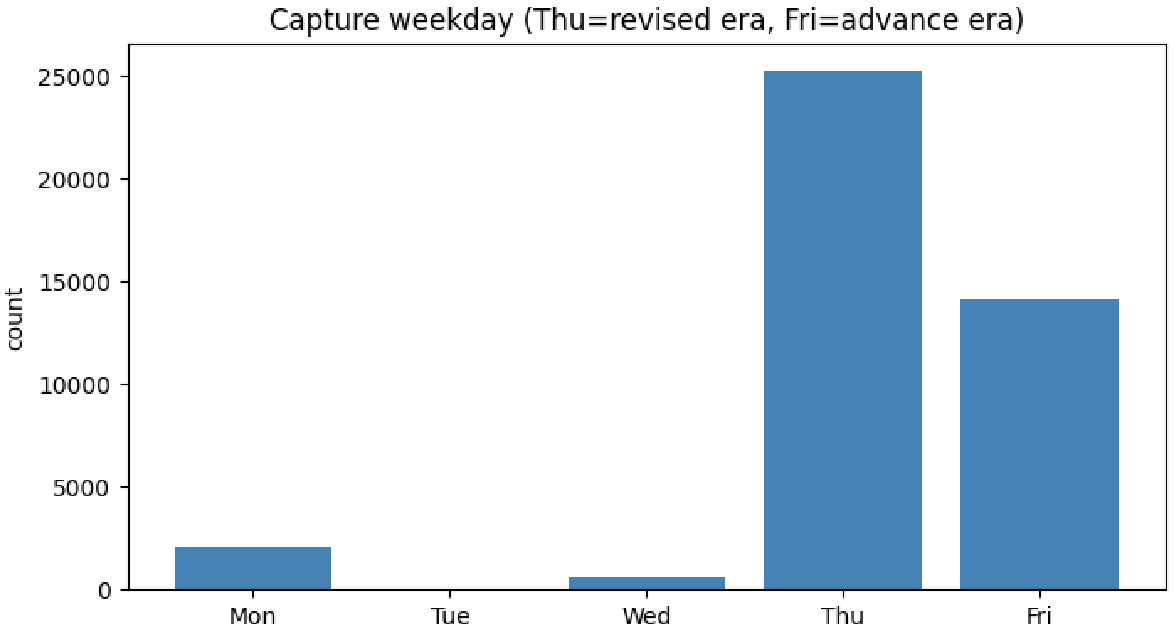
V3.7 — Has FRED's capture timing / stored print type been stable?

WARN

Criterion: stable median capture lag across the sample.

- **No — one structural break at ~2020-07-10.** Median pub_lag **19d before** (n=26,163) → **13d after** (n=15,961).
- Sat+19d = the following report Thursday → pre-2020 ALFRED stores the **revised print**. Sat+13d = the Friday after the advance report → post-2020 it stores the **advance print**.





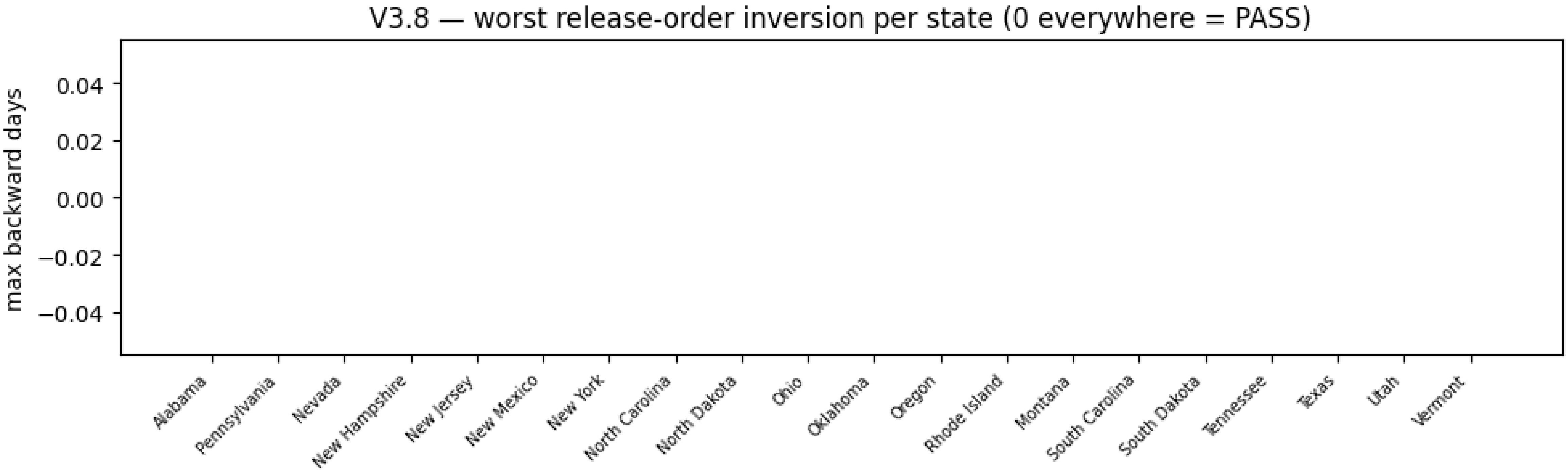
Note — One series, **two meanings**: pre-2020 values are one revision old; post-2020 values are preliminary. Backtests must use **per-era availability lags** (19d vs 13d), and V4.3 measures how much the print mixing distorts levels — spoiler: median |0.09%|, second-order.

V3.8 — Do weeks become available in order?

PASS

Criterion: 0 release-order inversions per state.

- **0 inversions** — in every state, first_release_date is weakly increasing in obs order; max backward step 0 days in all 51 states.



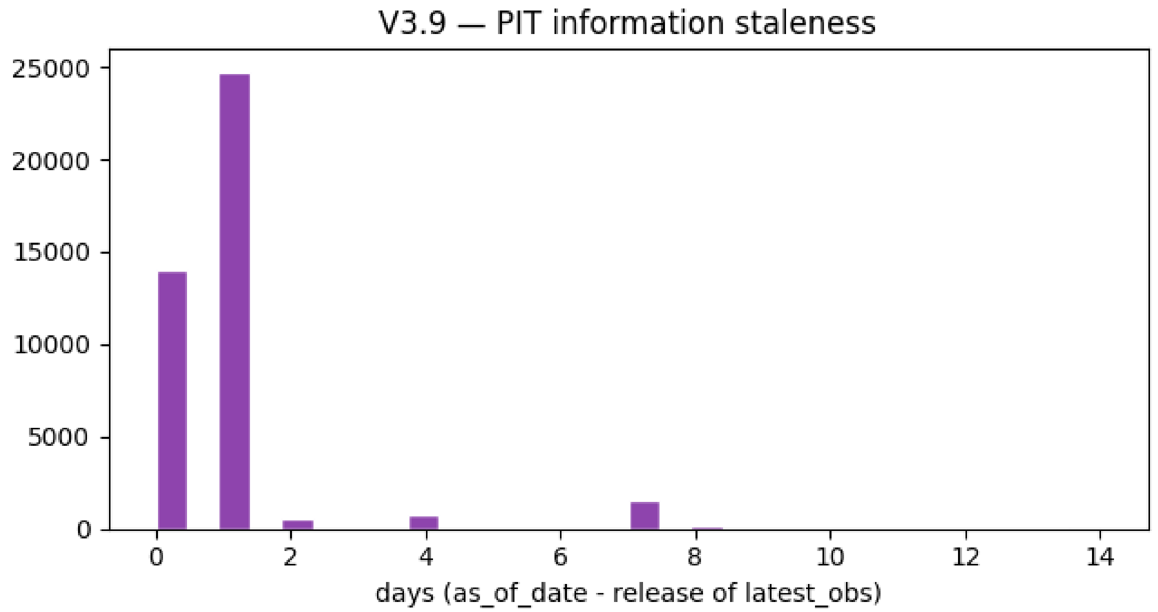
Note — An inversion would mean a later week became visible before an earlier one — poison for “latest available” logic. Monotonicity is what lets the pit builder use a simple cummax **availability frontier** + searchsorted, with no correction passes.

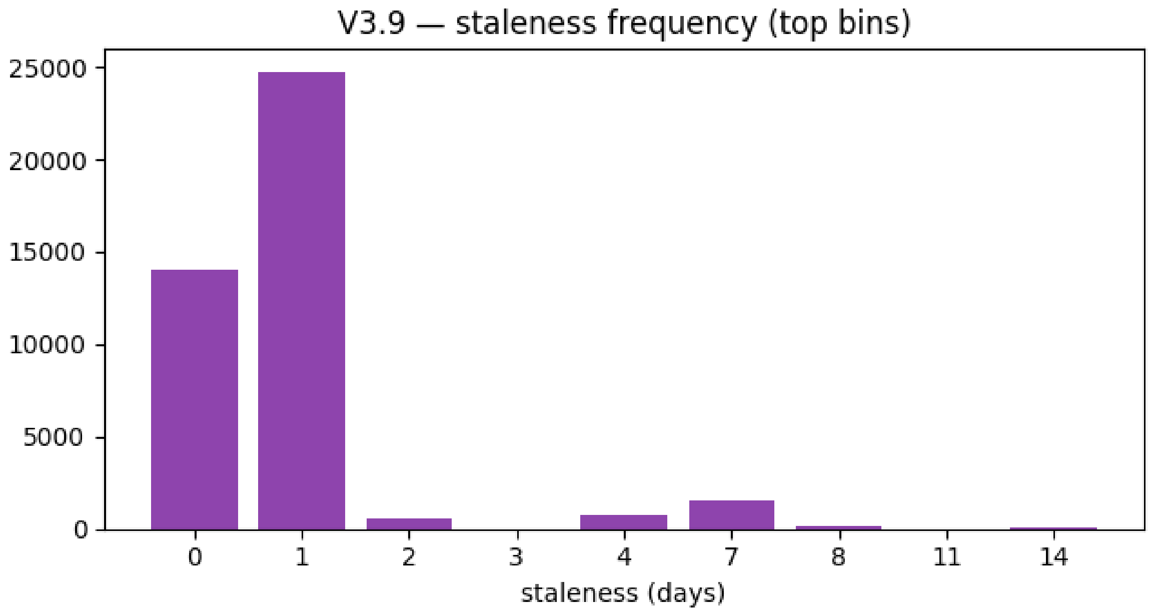
V3.9 — Does the PIT panel contain any look-ahead?

PASS

Criterion: 0 violations, 0 unmatched keys.

- Re-verified every one of **41,718 rows**: release date of the underlying value ≤ as_of_date — **0 violations, 0 unmatched keys**.
- Staleness (as_of – release of t0): **median 1 day, max 14 days** — the panel refreshes essentially the day after each capture.





Note — This is the hard gate for backtesting, run **against the vintage table, not against the builder's own bookkeeping** — an independent recomputation, so a builder bug could not certify itself.

MODULE

V4

First release and after: how often do the numbers change, where, by how much — and does the print mixing from V3.7 matter?

Revision behavior

check	question it answers
V4.1	How often does a first print get revised at all (endpoint: first vs current)?
V4.1b	Where are the revisions concentrated (states / periods)?
V4.2	Do values change mid-life and net out — invisible to the endpoint check?
V4.3	How different is an advance print from its revised print — does the era mixing matter?

V4.1 — How often does a first print get revised at all?

WARN

Criterion: PASS < 0.5% of observations revised; WARN < 5%.

- Endpoint join, first release vs current, all 42,124 CC obs: **1,403 revised = 3.33%** — an order of magnitude above the PASS bar.
- Among the revised: median **|0.31%|**, p75 3.7% — mostly small — but the tail reaches **13,505%** (data errors, next slide).
- **“Effectively unrevised” is a small-sample myth**: probe CA/TX/NY and you'd conclude ~0%; the full panel says otherwise.

```
  |revision| among the 1,403 revised obs (%)
p25    0.10      p50    0.31      p75    3.69
max    13,504.55   (Arizona 2026-01: 132 -> 17,958)
```

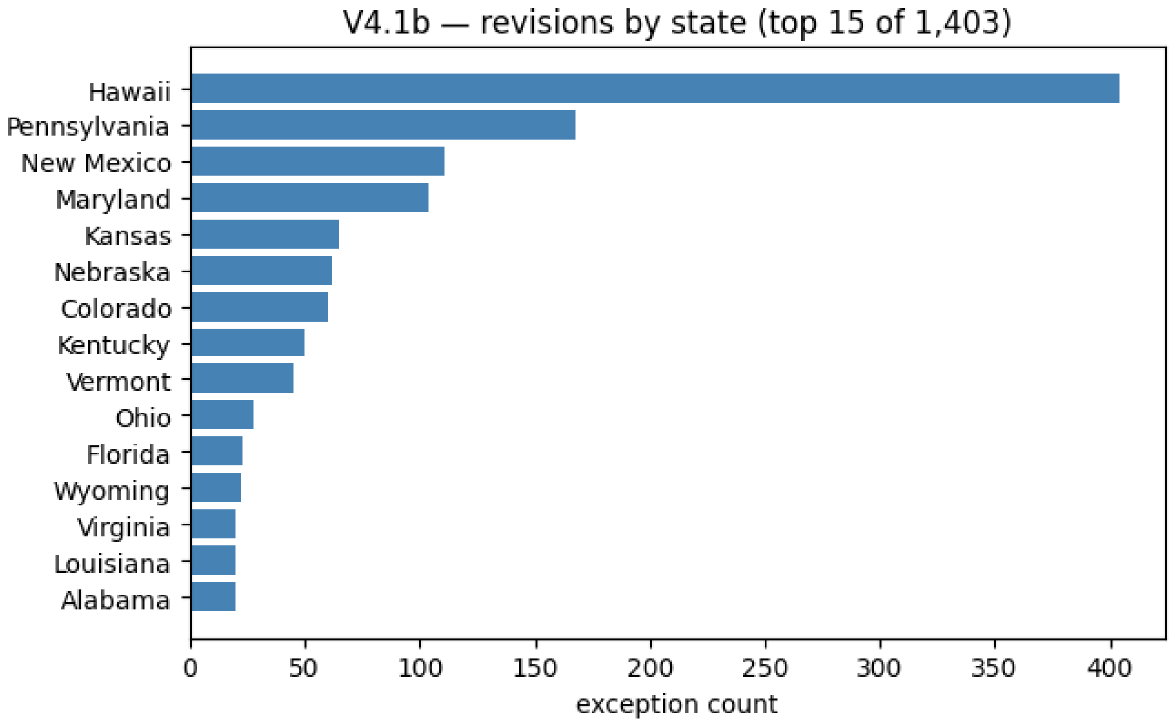
Note — Claims are still by far the least-revised of the three state labor datasets (U3 ≈86%, NFP ≈99%) — but 3.33% ≠ 0, and the exceptions cluster (V4.1b). The full exception list is saved to claims_revision_exceptions.csv and feeds the backtest-hygiene module.

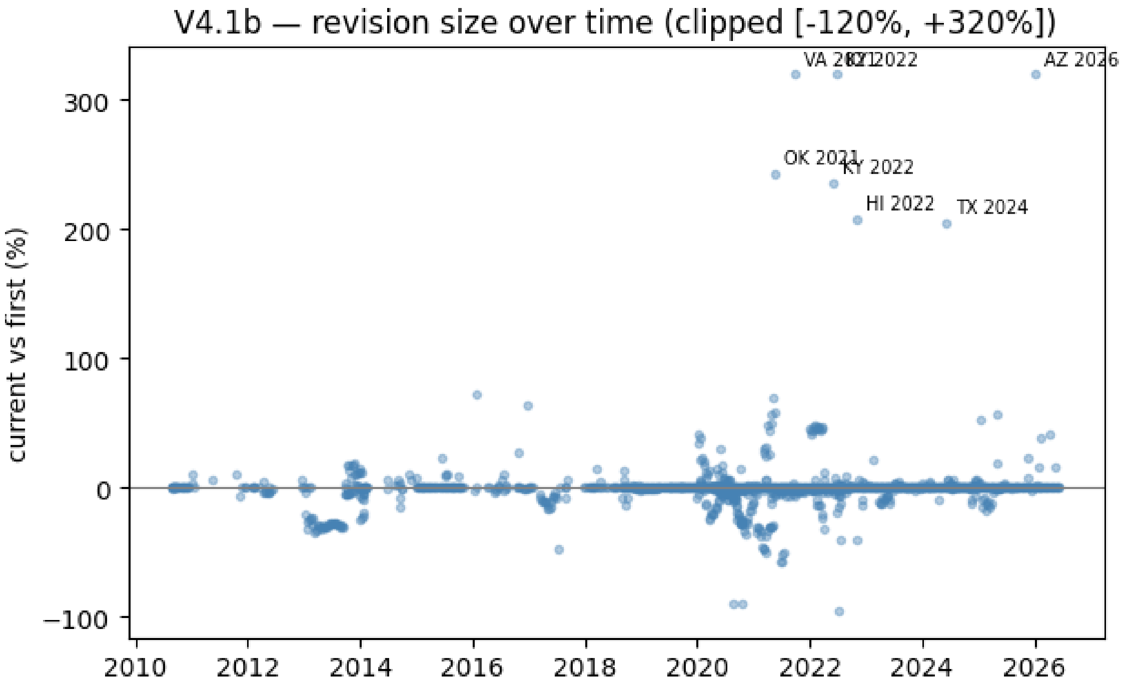
V4.1b — Where are the revisions concentrated?

WARN

Criterion: concentration documented (informational).

- **Top-4 states = 56%** of the 1,403 exceptions: Hawaii 404 (median |0.2%|), Pennsylvania 168 (|0.1%|), New Mexico 111 (|3.0%|), Maryland 104 (|0.1%|); by year, the bulge is **2020–22**.
- Largest: **AZ 2026-01: 132 → 17,958** (+13,505%) · **TX 2024-06: 48,748 → 148,748** (exactly +100,000 — a digit error) · CT / DC / AR ≈ 10× decimal slips in 2020–22.





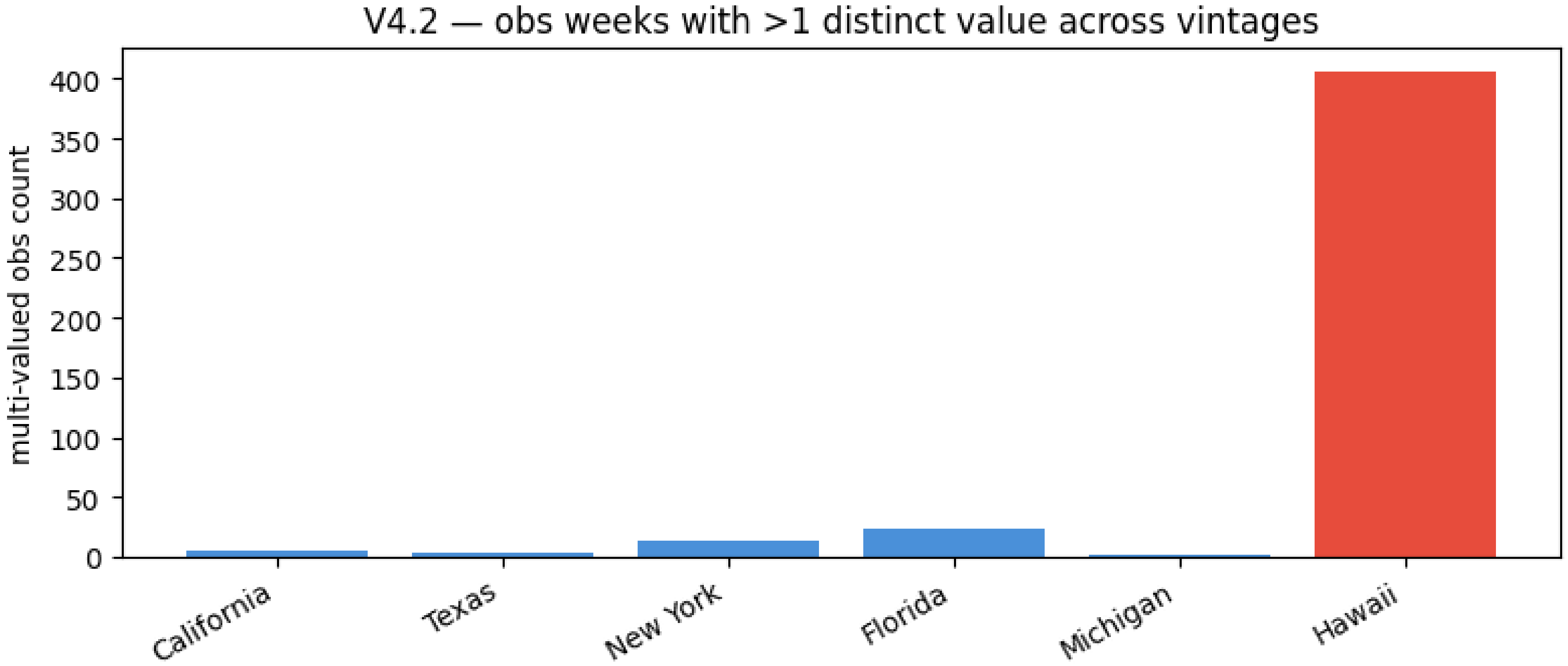
Note — Two different species here. **Chronic revisers** (HI, PA, MD): tiny, frequent, genuine re-reports — consider state-specific noise handling or a revised_later flag. **One-off data errors** (TX +100k, DC 10×): exclusion/dummy candidates — the first print was simply wrong, and a backtest fed the vintage value will trade on it.

V4.2 — Do values change mid-life and net out — invisible to the endpoint check?

PASS

Criterion: core states < 0.5% multi-valued obs (chronic states reported separately).

- Full vintage trajectories (every value a week ever had, output_type=2) for 6 states: core CA/TX/NY/FL/MI — **49 multi-valued obs of 10,556 = 0.46%**, never more than 2–3 values.
- **Hawaii: 405 of 2,152 obs (19%) carry up to 5 different values** across their life — its first prints are genuinely preliminary.



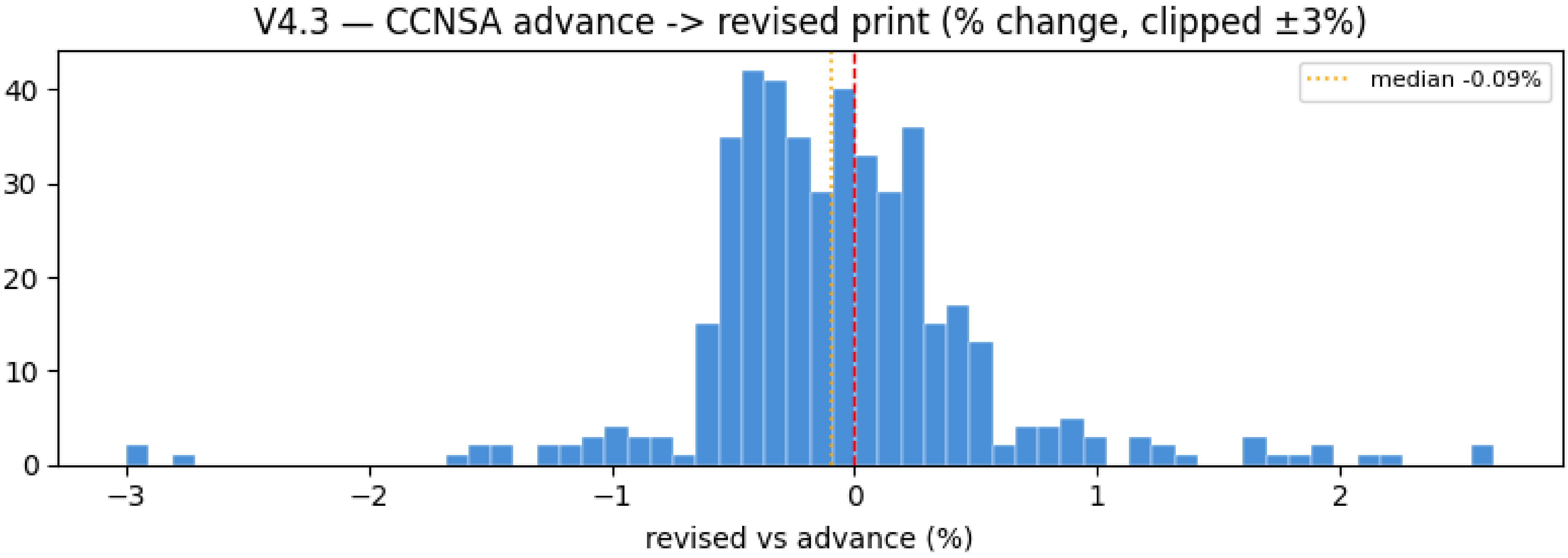
Note — For core states the cheap endpoint join (V4.1) captures the whole story — values don't churn mid-life and net out. Hawaii is the exception that justifies this check's existence: an endpoint comparison alone would undercount its churn.

V4.3 — How different is an advance print from its revised print — does the V3.7 era mixing matter?

PASS

Criterion: informational calibration on the national series.

- The national archive stores **both prints for 441 weeks**: advance → revised change **median –0.09%**, IQR [–0.39%, +0.21%], max **|6.4%|** (COVID weeks).
- The advance print **slightly overstates** claims on average — states trim their initial estimates more often than they raise them.



Note — VERDICT on V3.7's era mixing (revised prints pre-2020, advance prints post-2020): typical difference ~0.1%, tails ~6% only in COVID weeks — **second-order for state features**. Acceptable as-is; add an era dummy if precise levels matter.

Scorecard — every registered check

ID	check	result	headline evidence
V2.1	All obs_date are Saturdays (DOL week-ending)	**PASS**	0 non-Saturday rows (827 + 2,113 weeks)
V2.2a	Interior weekly gaps, pre-1990 startup era	**WARN**	48 missing weeks / 22 states, all 1986–88
V2.2b	Interior weekly gaps, post-1990	**PASS**	none
V2.3	Panel edges aligned across states	**PASS**	start spread 5w · end spread 1w
V3.1	ALFRED vintage boundary identified	**WARN**	PIT only from 2010-09-16 — pre-boundary is seasonality-only
V3.2	National vintages land mid-week	**PASS**	Thu 96.7% + holiday-Wed 3.3% of 867
V3.3	National first release = advance print at obs+12d	**PASS**	median = mode = 12d
V3.4	State captures cluster on release days	**PASS**	Thu 60.1% + Fri 33.5% + Wed 1.3%
V3.5	State capture vs national release windows	**PASS**	93.6% inside the two report windows
V3.5b	Era gap-pattern consistency	**WARN**	pre-2020 97.8%; post-2020 83.3% + 13% Monday slip
V3.6	No state value visible before its DOL report	**PASS**	min lag 13d ≥ 12d floor; 0 violations
V3.7	Capture-lag regime stability	**WARN**	break ~2020-07-10: median 19d → 13d (revised → advance print)
V3.8	Release dates weakly increasing per state	**PASS**	0 inversions
V3.9	PIT panel look-ahead guard	**PASS**	0 violations / 0 unmatched of 41,718 rows
V4.1	Endpoint revision rate	**WARN**	1,403 of 42,124 obs revised (3.33%)
V4.1b	Revision concentration	**WARN**	top-4 states = 56%; bulge 2020–22; TX +100k error
V4.2	Intermediate revisions bounded	**PASS**	core 0.46% multi-valued; Hawaii 19% (chronic)
V4.3	Advance vs revised print calibration	**PASS**	median –0.09%, IQR ±0.3%, max 6.4%

Module V1 uses print-style flags rather than registered rows: V1.1 OK · V1.2 OK · V1.3 OK · V1.4 WARN (15 pre-1990 zeros) · V1.4b WARN (6 pre-1990 interior zeros) · V1.5 OK.

What a model built on this data must respect

- **Index by release date, never by observation date.** Features live on the information clock; the pit panel (0 look-ahead, 0 holes, staleness ≤ 14d) is the only table a backtest may touch.
- **Two capture eras.** Before ~2020-07-10 the archive holds the *revised* print at Sat+19d; after, the *advance* print at Sat+13d. Use per-era availability lags; the print mixing itself is second-order (median |0.09%|, V4.3) — add an era dummy if levels matter.
- **The point-in-time world begins 2010-09-16.** Everything earlier (back to 1986) is current-only: seasonality and climatology, never backtests.
- **Revisions are rare but not mythical: 3.33%.** Concentrated in chronic revisers (HI, PA, NM, MD) and 2020–22; one-off data errors (TX +100,000; AZ 132→17,958) are exclusion candidates — the list ships in claims_revision_exceptions.csv.
- **Respect the gaps.** 48 pre-1990 missing weeks + 15 startup zeros: treat as NaN, never forward-fill a week-over-week change across them. Post-1990 the panel is hole-free.
- **Weekly cadence is trustworthy.** Saturdays 100%, national Thursdays 96.7% (+ holiday Wednesdays), monotone release order, balanced 51×818 pit grid — the mechanics all check out.

Note — Verdict: **12 PASS · 6 WARN · 0 FAIL** — cleared for forecasting, lead-lag and backtesting under the rules above. The WARNs are documented properties of the data's plumbing, not defects; the deeper modules (outliers, cross-sectional coherence, seasonality, lead-lag orientation, backtest hygiene, governance) run in the full suite claims_validation.ipynb as V5–V10.